

UNIT-3

MACHINE LEARNING

CONVOLUTIONAL NEURAL NETWORKS (CNNs) ²⁰²⁴

A Convolutional Neural Network (CNN) is a special type of artificial neural network used mainly for processing images, videos and visual data.

It is designed to automatically detect important features such as edges, shapes, textures, and objects from input ~~do~~ images.

It is one of the most important technique in Deep learning and computer vision.

Main Idea of CNN

A normal neural network treats every pixel separately, but CNN processes images using small filters called kernels.

- These filters move across the image and extract useful information.

FLATTENING in CNN

Flattening is the process of converting a multi-dimensional features map (2D or 3D data) into a single one-dimensional vector.

It is used in a Convolutional Neural Network before sending data to the fully connected layer.

Why Flattening is Needed

After convolution and pooling operations, the output is still in matrix form.

- Features map size = $4 \times 4 \times 3$

But the fully connected layer accepts only a 1D vector.

So flattening converts.

- $4 \times 4 \times 3 \rightarrow 48$ values in one row.

Example $\rightarrow$ Suppose pooling layer output is

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

After Flattening

$$[1, 2, 3, 4]$$

Important Points

- Converts 2D/3D data into 1D data
- Used between pooling layer and fully connected layer
- Does not lose data values
- Makes data ready for classification.

Advantage

- Simple conversion process
- Help connect CNN layer to dense layer
- Easy classification

Disadvantage

- Large feature maps create very large vectors
- Can increase computations.

Role in CNN $\rightarrow$ CNN layers extract features. while flattening prepares those features for final classification

Subsampling in CNN

Subsampling is the process of reducing the size of feature map in deep learning on MC while keeping the important information.

In CNN, subsampling is mainly done using the pooling layer.

It reduces

- Width
 - Height
 - Number of computations.
- without losing important features.

Purpose of Subsampling

- Reduce image dimensions
- Reduce computation time
- Prevent overfitting
- Keep important features only

- Subsampling is also known as pooling
- The main goal of subsampling is to make the network more efficiently by reducing the amount of data that must be processed in later layers.

Types of subsampling.

① Max Pooling.

Max pooling selects the largest value from each pooling region

for the example above.

$$\max(1, 3, 5, 6) = 6$$

$$\max(2, 4, 1, 2) = 4$$

$$\max(7, 8, 1, 2) = 8$$

$$\max(3, 4, 5, 6) = 6$$

Output $\rightarrow$ $\begin{matrix} 6 & 4 \\ 8 & 6 \end{matrix}$

Advantages

- Preserves the strongest features.
- Works well for image recognition tasks

② Average Pooling.

Average Pooling computes the average value within each pooling region.

Example

$$\rightarrow (1 + 3 + 5 + 6) / 4 = 3.75$$

Output $\rightarrow 3.75$

Advantages:- Retains more background information.

- Produces smoother feature maps.

Padding 2025, 2024

- Padding is a technique used in CNN, where extra pixels (usually zeros) are added around the border of an input image or feature map before applying a convolution operation.
- Padding helps control the size of the output feature map and ensures that important ^{information} near the image boundaries is not lost.

Why is Padding Needed.

Without padding, the size of the feature map decreases after every convolution.

For example:-

Input image size:- 5×5

kernel size:- 3×3

stride:- 1

$$\text{Output size} = \frac{(5-3)}{1} + 1 = 3$$

$$\text{Output} = 3 \times 3$$

★ The feature map becomes smaller. After several convolution layers, the image may shrink too much.

Padding solves this problem by adding extra pixels around the image.

Example of Padding (original image) 5×5

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

After Adding Padding = 1

```
00000000
01111110
01111110
01111110
01111110
01111110
00000000
```

The image size becomes $\rightarrow 7 \times 7$.

Stride

Stride is the number of pixels by which the convolution filter (kernel) moves across the input image at each step.

Example

- stride = 1 $\rightarrow$ The filter moves 1 pixels at a time.
- stride = 2 :- The filter moves 2 pixels at a time, producing a smaller output feature map.

Effects of stride.

- Smaller stride (1) :- Captures more details and produces a larger output.

- Larger stride (2 or more) :- Reduce computation and produces a smaller output.

In short

- "stride is the number of pixels the convolution kernel moves across the input image during the convolution operation."

Convolutional Layer in CNN

A convolutional layer is the core building block of a convolutional Neural Network CNN.

It is responsible for extracting important features such as edge, textures, shapes, and patterns from the input image.

A convolutional layer applies a set of small filters (kernels) to the input image.

Working of a convolutional layer.

1. Input Image. - The image is provided as input to the convolutional layer.

2. Kernel (Filter)

- A small matrix (e.g. 3×3 or 5×5) is used to detect specific features.

3. Convolution Operation.

- The filter moves across the images
- At each position, element-wise multiplication is performed between the filter and the image region.
- The results are summed to produce a single value.

4. Features Map

- The output generated after applying the filter is called a feature map.

Components of a convolutional layer.

- Input :- Image or feature map from the previous layer.
- Kernel (Filter) - Detects patterns and features.
- Stride - Determines how many pixels the filter moves each step.
- Padding :- Adds extra pixels around the image to preserve dimensions.
- Activation function :- Introduces non-linearity.

Advantages.

- Reduces the no. of parameters compared to fully connected layers.
- Automatically extracts features
- Improves image recognition performance

Pooling Layer in CNN 2025

Pooling layer is a layer in a CNN that reduces the size of feature maps while preserving the most important information.

It helps decrease computation memory usage and overfitting.

The pooling layer performs a downsampling operation on the feature map generated by the convolutional layer.

Working of Pooling layer.

1. The feature map is divided into small regions (e.g. 2×2)
2. A pooling operation is applied to each region.
3. One value is selected from each ~~each~~ region to create a smaller feature map.

Advantages of Pooling

- Reduces feature map dimensions
- Decreases computational cost
- Reduces overfitting
- Makes the model more robust to small shifts and distortions.

Disadvantages.

- Some information may be lost during downsampling.
- Excessive pooling can reduce important details

Loss Layer

Loss layer is the final layer in CNN. that calculate the error (loss) between the model's predicted output and the actual target output.

The loss layer measures how far the predicted result is from the correct result. The objective of training is to minimize this loss.

Working of a Loss Layer.

1. The CNN produces prediction
2. The prediction is compared with the actual target value.
3. A loss function computes the error
4. The error is propagated backward through the network.
5. Weights are updated to reduce the loss.

Common Loss Functions.

1. Cross-Entropy Loss:- Used for classification problems.

For example, Actual class : cat

Predicted probability :- cat = 0.9
Dog = 0.1

A higher probability for the higher correct class results in lower loss.

2. Mean Squared Error (MSE)

Used mainly for regression tasks.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where.

y_i = Actual value

$\hat{y}_i$ = Predicted value

n = Number of samples.

Importance of the loss layer.

- Evaluates model performance
- Guides the learning process.
- Helps optimize network parameters

Inception Network. 2024

The Inception Network, also known as GoogLeNet, is a deep CNN architecture developed by researchers at Google.

It won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2014.

An Inception Network in CNN architecture that performs multiple convolution operations of different sizes (1×1 , 3×3 and 5×5) in parallel and then combines their outputs.

This allows the network to capture features at different scales.

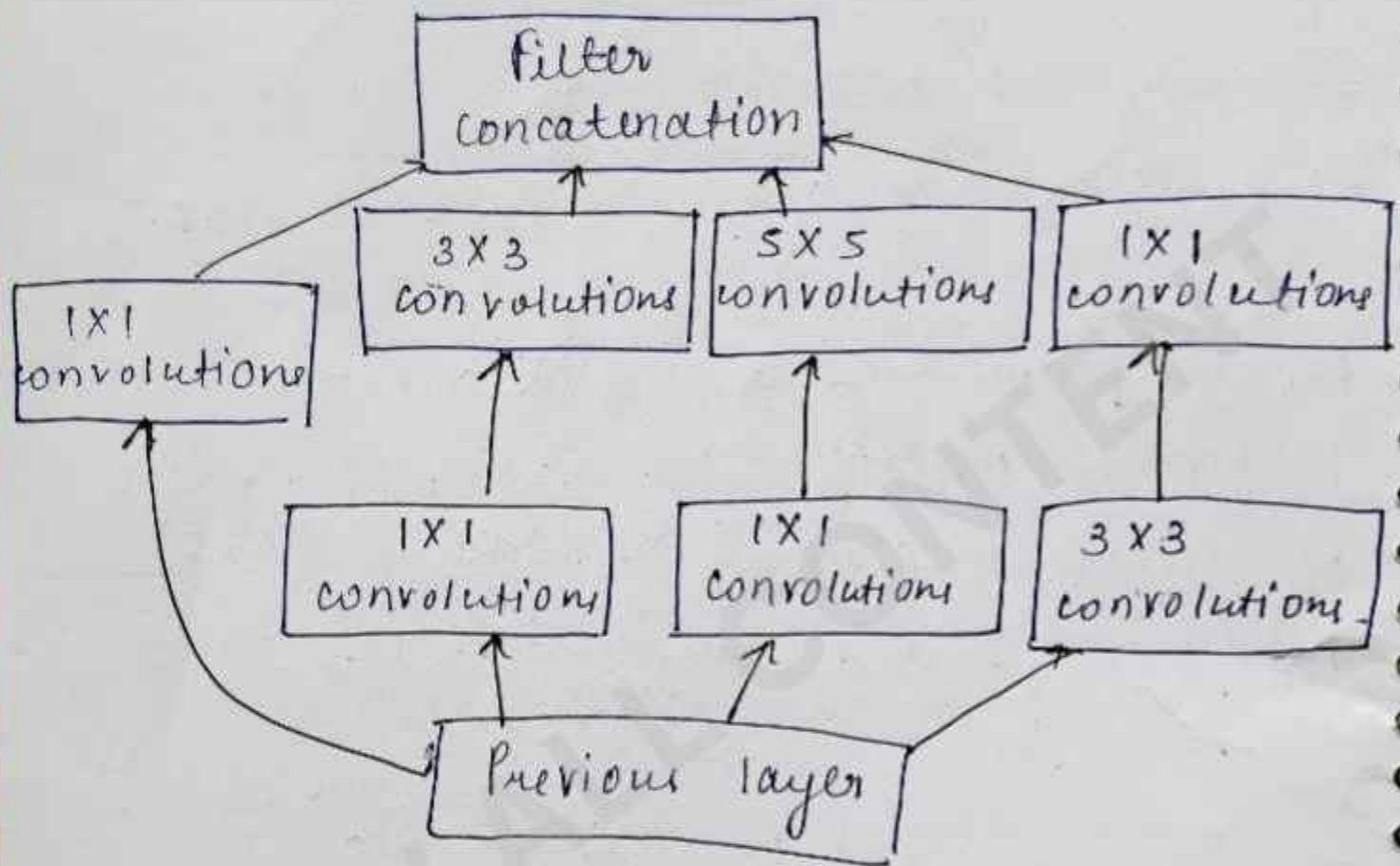
Inception Module.

The main building block of GoogLeNet is the Inception Module.

It contains:

- 1×1 Convolution
- 3×3 Convolution
- 5×5 Convolution
- 3×3 Max pooling.

All these operations are performed in parallel, and their outputs are concatenated into a single feature map.



Inception Network.

Role of 1×1 convolution

The 1×1 convolution is used before 3×3 and 5×5 convolutions to :-

- Reduces the number of channels
- Lower computational cost
- Reduce memory usage

Transfer Learning.

Transfer Learning is a ML technique in which a model trained on one task is reused for another related task. Instead of training a new model from scratch, the knowledge learned by a pre-trained model is transferred to the new problem.

Working.

1. Train a model on a large dataset (source task).
2. Save the learned features and weights.
3. Use the pre-trained model for a new related task (target task).
4. Retrain only the final layers or fine-tune the model.

Types.

- ① Feature Extraction: Freeze pre-trained layers and trained only the new output layers.

②. Fine-Tuning - Retrain some of the pre-trained layers along with the new layers.

Advantages.

1. Requires less training data
2. Reduce training time
3. Improves accuracy
4. Reduces computational cost.

Disadvantages.

1. Not effective if source and target tasks are very different
2. Choosing the wrong pre-trained model may reduce performance

Application

- Image Classification
- Object Detection
- Face Recognition
- Medical Image Analysis.

One Shot Learning.

One shot learning is a technique in which a model learns to recognize or classify an object from only one training example.

Unlike traditional ML, which requires many examples, One-Shot Learning can learn from a very small amount of data.

Working.

1. The model is trained to compare similarities between objects.
2. Given one example of a new class, it learns its features.
3. When a new sample arrives, the model compares it with the stored example.
4. If they are similar, it assigns the same class.

Example → Suppose a face recognition system sees only one photo of a person.

Later, when another photo of that person is shown, the system identifies the person by comparing facial features with the single stored image.

Application

- Face Recognition
- Signature Verification
- Object Recognition.
- security System.

Dimension Reductions. 2025

Dimension Reduction is the process of reducing the number of input features (variables) in a dataset while keeping as much important information as possible

for example :-

Student	Age	Height	Weight	Marks	Attendance
A	18	176	65	85	90
B	19	168	62	80	88

Here, Age, Height, Weight, Marks and Attendance are 5 dimensions (features)

If some features are unnecessary or highly related, we can reduce them to fewer features. This process is called Dimension Reduction.

Need of Dimension Reduction.

Datasets in ML may contain many features

• Too many features can:-

- Increase Training time
- Require more memory
- Cause Overfitting
- Make Visualization difficult.

• Dimension Reduction helps simplify the data.

Types of Dimension Reduction.

- ① Feature selection.
Select only the most important features and remove the rest.

Example:- From . . Age, Height, weight, marks, attendance.

keep → Attendance, marks, Age.

Remove less useful features.

Methods → Filter methods

- Wrapper methods
- Embedded methods.

2. Feature Extraction.

Create new features by combining existing features.

Example: Instead of using

- Math Marks
- Physics Marks
- Chemistry Marks.

Create:

- Overall Performance Score.

Advantages.

1. Training is faster
2. Less storage required
3. Reduce overfitting
4. Better model performance
5. Easier data visualization.

Disadvantages.

1. Some information may be lost
2. Reduce interpretability.
3. Choosing the wrong technique can decrease accuracy.

Implementation of CNN like tensor flow

2024

A CNN is a deep learning model widely used for image classification, object detection, and computer vision tasks. TensorFlow provides the Keras API to easily build, train and evaluate CNN models.

1. Import Required Libraries.

First import TensorFlow and Keras libraries required for building the CNN model.

Python

```
import tensorflow as tf
from tensorflow.keras import
datasets, layers, models
```

Purpose - Provides tools for data handling, model creation, training and evaluation.

2. Load the Dataset

Load the training and testing data.

Purpose - The dataset contains input images and their corresponding labels.

3. Data Preprocessing

Normalize pixel values from 0-255 to 0-1

Purpose : • Improves training speed

- Helps the model converge faster.
- Reduces numerical instability

4. Build the CNN Architecture.

Create the CNN model using convolution, pooling and dense layer

5. Compile the model

Specify optimizer, loss function, and evaluation metric.

Parameters.

- Optimizer - updates weights efficiently
- Loss function - measures prediction
- Accuracy - Evaluate classification performance

6. Train the Model

Train the CNN using the training dataset.

7. Evaluate the Model

~~Train~~ Test the trained model on unseen data.

Purpose.

- Measures model accuracy
- Determines generalization performance.

8. Make Predictions.

- Used the trained model to classify new images.
- The class with the highest probability is selected as the final prediction.

Training Process Flow Diagram.

